

Getting Started

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Instructions

This document runs a simple analysis of the Table 1 from (Wansink and Payne 2009). Edit the header information to show your name and the date you complete the assignment.

Typeset this document in RStudio. You may need to install additional Python libraries; you can run the code in comments (i.e.)

```
python3 -m pip install pandas
```

from the command line to install the necessary libraries. On some systems, `python` may be used instead of `python3`

When you are able to typeset the original document on your computer, modify the code to analyze Calories per Serving. That is, replace all occurrences of the variable names of the form `CaloriesPerServing*` to the variable `CaloriesPerServing*`. Change the values in the table named `ReferenceValues` to match the values in the text of Wansink and Payne (Wansink and Payne 2009).

Typeset the edited document and analyze the results. Add your own comments to the section headed **Comments**. Format your comments as *italicized* text, **bold** text or

as quoted text.

Change the name of this file to match your user name on D2L, keeping the ‘Rmd’ extension, and include week number and coding language in the title (for example, `Peter.Claussen.Python.1.Rmd`). Upload this file to D2L. Typeset this file to Word or PDF and upload the result to D2L as well.

Data

Table 1: Mean and (SD) for selected recipes from “Joy of Cooking”

Measure	1936	1946	1951	1963	1975	1997	2006
calories per recipe (SD)	2123.8 (1050.0)	2122.3 (1002.3)	2089.9 (1009.6)	2250.0 (1078.6)	2234.2 (1089.2)	2249.6 (1094.8)	3051.9 (1496.2)
calories per serving (SD)	268.1 (124.8)	271.1 (124.2)	280.9 (116.2)	294.7 (117.7)	285.6 (118.3)	288.6 (122.0)	384.4 (168.3)
servings per recipe (SD)	12.9 (13.3)	12.9 (13.3)	13.0 (14.5)	12.7 (14.6)	12.4 (14.3)	12.4 (14.3)	12.7 (13.0)

Analysis

Enter data

```
#python -m pip install pandas
#library(reticulate)
#py_install("pandas")
import pandas
CookingTooMuch = pandas.DataFrame([[1936, 2123.8, 1050.0, 268.1, 124.8, 12.9, 13.3],
                                   [1946, 2122.3, 1002.3, 271.1, 124.2, 12.9, 13.3],
                                   [1951, 2089.9, 1009.6, 280.9, 116.2, 13.0, 14.5],
                                   [1963, 2250.0, 1078.6, 294.7, 117.7, 12.7, 14.6],
                                   [1975, 2234.2, 1089.2, 285.6, 118.3, 12.4, 14.3],
                                   [1997, 2249.6, 1094.8, 288.6, 122.0, 12.4, 14.3],
                                   [2006, 3051.9, 1496.2, 384.4, 168.3, 12.7, 13.0]],
                                  columns=['Year', 'CaloriesPerServingMean1', 'CaloriesPerServingSD1', 'CaloriesPerServingMean2',
                                           'CaloriesPerServingSD2', 'ServingsPerRecipeMean', 'ServingsPerRecipeSD']
)
```

Create values for confidence interval plot

Wansink reports that 18 recipes were analyzed.

```
n = 18.0
```

Use standard formula for standard error $\widehat{se} = \widehat{\sigma}/\sqrt{n}$ and confidence interval, assuming a significance level α of 5%.

$$CI(95\%) = \widehat{\mu} \pm z_{\alpha/2} \times \widehat{se}$$

```
# python3 -m pip install scipy
import scipy
import math
def StandardError(sigma, n):
    return sigma/math.sqrt(n)

def ConfidenceBound(sigma, n, alpha = 0.05):
    return scipy.stats.norm.ppf(1-alpha/2)*StandardError(sigma,n)
```

Create a variable for plotting and calculate upper and lower bounds using confidence intervals.

```
PlotCookingTooMuch = CookingTooMuch
PlotCookingTooMuch['CaloriesPerServing'] = PlotCookingTooMuch['CaloriesPerServingMean1']
PlotCookingTooMuch['Lower'] = PlotCookingTooMuch['CaloriesPerServing'] - ConfidenceBound(CookingTooMuch['CaloriesPerServingSD1'], n)
PlotCookingTooMuch['Upper'] = PlotCookingTooMuch['CaloriesPerServing'] + ConfidenceBound(CookingTooMuch['CaloriesPerServingSD1'], n)
PlotCookingTooMuch['Bound'] = ConfidenceBound(CookingTooMuch['CaloriesPerServingSD1'], n)
PlotCookingTooMuch = PlotCookingTooMuch[["Year", "CaloriesPerServing", "Lower", "Upper", "Bound"]]
```

Examine the values to make sure we've entered correctly.

```
print(PlotCookingTooMuch)
```

	Year	CaloriesPerServing	Lower	Upper	Bound
## 0	1936	2123.8	1638.733661	2181.453599	485.066339
## 1	1946	2122.3	1659.269532	2179.676418	463.030468
## 2	1951	2089.9	1623.497166	2143.580675	466.402834
## 3	1963	2250.0	1751.721378	2304.373627	498.278622
## 4	1975	2234.2	1731.024518	2288.850807	503.175482
## 5	1997	2249.6	1743.837498	2305.960089	505.762502
## 6	2006	3051.9	2360.703566	3129.649205	691.196434

Check our calculations. Wansink reports 95% confidence intervals for the 1936 and 2006 means as [210.4, 325.8] and [325.8, 514.7], respectively. We should be equal, to within rounding error.

```
# python3 -m pip install numpy
import numpy

CompValues = PlotCookingTooMuch.loc[[0,6]]
CompValues=CompValues[["Lower", "Upper"]]

ReferenceValues = numpy.matrix([[210.4, 325.8], [325.8, 514.7]])
CompValues
```

	Lower	Upper
## 0	1638.733661	2181.453599
## 6	2360.703566	3129.649205

```
ReferenceValues
```

```
## matrix([[210.4, 325.8],
##         [325.8, 514.7]])
```

```
Comparisons = abs(CompValues-ReferenceValues)>0.1
Comparisons.any(axis=None)
```

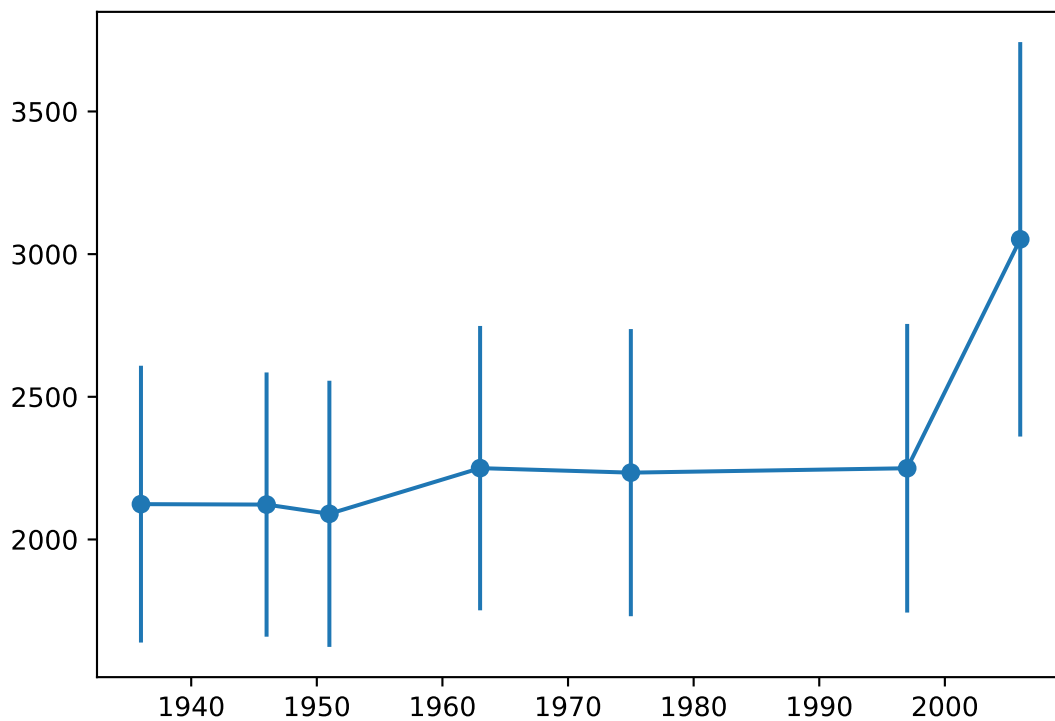
```
## True
```

We no longer need the original data.

```
CookingTooMuch = None
```

Plot the table

```
# python3 -m pip install matplotlib
import matplotlib.pyplot
plt = matplotlib.pyplot.scatter(PlotCookingTooMuch['Year'], PlotCookingTooMuch['CaloriesPerServing'])
plt = matplotlib.pyplot.errorbar(PlotCookingTooMuch['Year'], PlotCookingTooMuch['CaloriesPerServing'],
                                yerr=PlotCookingTooMuch['Bound'])
matplotlib.pyplot.show()
```



Comments

From this plot, it appears that average calories per *serving* changed very little from 1936 to 1997. This suggests that the significant increase from 1936 to 2006 reported by Wansink is due to an increase in the

2006 recipes only. Thus, we only need to compare 1936 and 2006 recipe sets.

References

Wansink, Brian, and Collin Payne. 2009. “The Joy of Cooking Too Much: 70 Years of Calorie Increases in Classic Recipes” 150 (150): 291–92.